

FreeBSD and Linux. I never have a reason to use Microsoft platforms for myself. So the transition from home to work and back again was nearly seamless.

I can't compare the ease of installation of VMware Player on the basis of this experience, because installation was done for me. But my sense was that VMware Player was a lot slicker than VirtualBox. It just seemed to work; it didn't have the USB restrictions and it felt more comfortable. It may be something to do with my comfort zones. Bear in mind that in the previous exercise, VirtualBox was a Windows application; in this case, VMware Player was a Linux application. It would be fair to compare either Windows VirtualBox with Windows VMware; or the Linux variants of both.

The third time: MS Windows/VMware Player/Linux/FreeBSD

I now work for a university. When I arrived, my desktop machine was a fairly standard Intel Core i7 running XP.

I don't usually like to talk too much about hardware. The only reason I mention this at all is because it becomes relevant shortly. All the work machines I've used have been much more powerful than anything that I have at home. From memory, the laptop of the first section was a 2 GB dual-core. The PC in the second section was either 2 GB or 4 GB, and also dual-core. My current PC is a 4 GB i7 (4 cores hyper-threaded). You would expect it to leave the earlier machines in its wake—and, for most things, it does.

Of course, I wasn't happy with my lot. This time I installed VMware Player. I did a little research. I learnt one feature relevant to VMware: you can go to places like <http://www.thoughtpolice.co.uk/> and download a pre-built VM which VMware can play. So you can choose to build your own, or download one that someone prepared earlier.

Again, I've forgotten some of the details. But, fortunately, I have notes from back then:

I guess I'm building infrastructure. I have a couple of VMs running. Why two? Well the first one has a windowing system (good) but it's a flavour of FreeBSD (neutral) and certain things are just looking too hard (really bad). I didn't install it. I downloaded a pre-built VM from a site that does these things.

I then downloaded a pre-built CentOS machine (good), but it was configured as a server (neutral) which meant that it had no windowing system (bad). Since I already had the BSD VM, I decided to use it for its windows to run xterms and SSH into the CentOS VM (pretty cool). If I have to live with a Windows PC, I'll invest the time to install a windowing system on the CentOS VM, and ditch the BSD VM.

For the moment, I'm really only using the CentOS VM to confirm usability.

A week later, I had made my first entry in the CentOS VM's History file (see Sidebar). My other notes said:

I have a lot more infrastructure in place. My CentOS

VM is actually not too bad. I live there about 40 per cent of the time. I don't think I need the BSD VM any more.

And a week after that:

Infrastructure is now in a state such that I think I'll work this way until the Christmas break. During the break, I might try to switch over (run Linux on the metal, Windows in a VM).

History files

Back in the days when I used to work for a small company called Optimization, I was introduced to the concept of history files. Every machine had a file, `/History`, in which changes were recorded. (For a non-UNIX machine, we would use the nearest equivalent, for example `C:\History.txt`.) Since Optimization was a small company, there was no splitting into 'silos'; on the contrary, everyone did everything. The idea was that if you performed some systems administration on a machine, you recorded that fact in an entry in the machine's history file. Then, if someone came along and discovered a problem, he could browse the recent entries in the history file to see if they provided a clue to the cause of the problem.

I was so taken by the concept, that I have adopted it in almost everything I do around computers. All my machines at home have history files; and as far as possible, wherever I have worked, I have maintained history files.

Lately, I have modified the concept slightly. I now have a history directory, `/History_dir`, under which I keep one or several history files, often under RCS version control. Why several history files?

This article ought to give you a clue. If I'm running any VMs, each one gets a separate history file. Although it might make sense to maintain the history file of a VM in `/History` in the VM, there are times when I would like to be able to read the history file of a VM without having to fire it up.

Evaluation

The pre-built VM was CentOS 5.5. By this time, I had learnt that many of the university's servers ran Red Hat. By running CentOS, I could develop scripts and run software in a very similar environment. CentOS gave me something else: a very conservative platform. By the time software migrates from Fedora to Red Hat to CentOS, it has had many user operating hours. For me, stability and reliability were far more important than anything I might get from having the latest releases.

VMware Player was very easy to install, and very easy to use. I now had a like-for-like comparison with the first case above (MS Windows VirtualBox and MS Windows VMware).

It was not my intention, when writing this article, to provide extensive comparisons between one VM environment and another. I simply wanted to describe various experiences and indicate some of the options available; there are many others. For a recent review, see [2] in *References*.